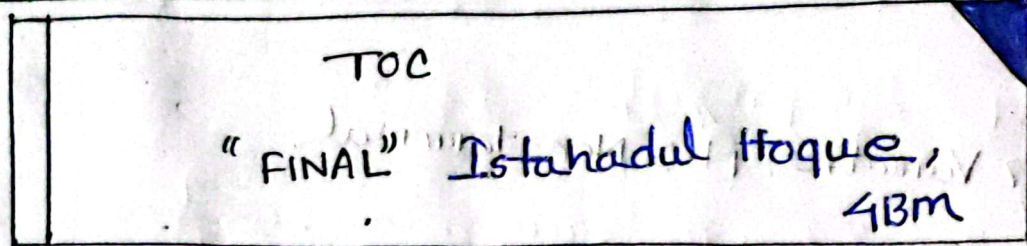


~~Construct CFG for set~~

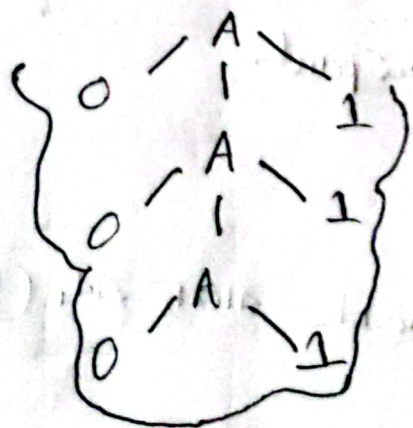


Context free Grammar

* A CFG is a set of rules/productions used to generate patterns of strings.

$A \rightarrow 01 \mid 0A1$ or, $A \rightarrow 01$

$A \rightarrow 0A1 = \{01, 001, 011, 0011, 0111, 00111, \dots\}$



parse tree

$01 \quad n=1, 0^1 1^1 = 01$

$0011 \quad n=2, 0^2 1^2 = 0011$

$000111 \quad n=3, 0^3 1^3 = 000111$

A context-free grammar is a 4-tuple (V, T, R, S)
 $N, \Sigma, P.$

① $V \rightarrow$ Variable, $N \rightarrow$ Non-terminal

② $T \rightarrow$ Terminal, Σ

③ $R \rightarrow$ Rules, $P \rightarrow$ Productions.

④ $S \rightarrow$ starting variable

$S \in V$

Ex:-

<u>LHS</u>	<u>RHS</u>
$A \rightarrow$	$0A1$
$A \rightarrow$	01
$B \rightarrow$	$\#$

Variable = $\{A, B\}$

$\rightarrow$ in L.H. Side

Terminal = $\{0, 1, \#\}$

$\rightarrow$ variable baad aa, (RHS)

starting variable = A

Identify the terminals, non-terminals, start variable, from given CFG:-

$$\textcircled{1} E \rightarrow E+T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow (E) \mid \text{id}$$

$$\text{Non-terminal} = \{E, T, F\}$$

$$\text{Terminal} = \{+, *, '(', ')', \text{id}\}$$

$$\text{Starting variable} = \{S\}$$

$$\textcircled{2} S \rightarrow (L) \mid a.$$

$$L \rightarrow L S \mid S.$$

$$\text{Non-terminal} = \{S, L\}$$

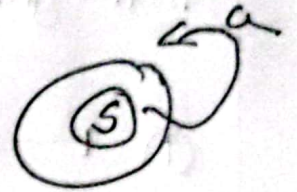
$$\text{Terminal} = \{ '(', ')', 'a', ',' \}$$

$$\text{start variable} = \{S\}$$

Design CFG

① $L = \{a^n \mid n \geq 0\}$.

$= \{\epsilon, a, aa, aaa, aaaa, \dots\}$



$S \rightarrow aS$

$S \rightarrow \epsilon$

a^*
 $= \{\epsilon, a, aa\}$

(0 or more 'a' characters)

CFG: $S \rightarrow aS \mid \epsilon$

$\boxed{\because n \geq 0}$

generate :-

"aaaa"

$S \rightarrow aS$

$\rightarrow aaS$

$\rightarrow aaas$

$\rightarrow aaaaS$

$\rightarrow aaaa\epsilon$

$\rightarrow aaaa$

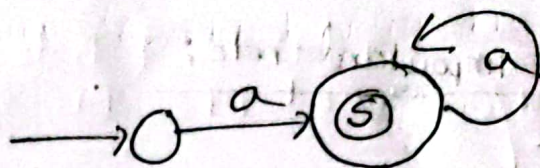
② $L = \{a^n \mid n \geq 1\}$

$= \{a, aa, aaa, \dots\}$

$S \rightarrow aS \mid a$

$$\begin{array}{l} S \rightarrow aS \\ S \rightarrow a \end{array}$$

generate: "aaa"

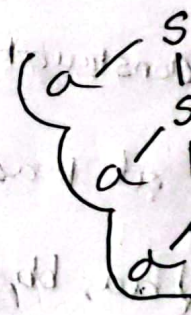


$S \rightarrow aS$

$\rightarrow aaaS$

$\rightarrow aaaa$

parse tree:



③ $L = \{\text{set of all strings over } a \text{ and } b\}$

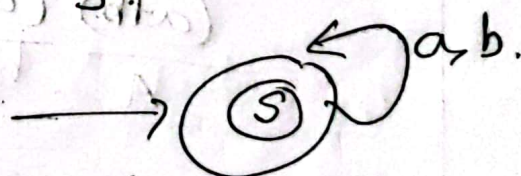
$= \{\epsilon, a, b, aa, bb, ab, ba, \dots\}$

CFG: $S \rightarrow aS \mid bS \mid \epsilon$

generate: bbab

$$\begin{array}{l} S \rightarrow aS \\ S \rightarrow bS \\ S \rightarrow \epsilon \end{array}$$

$(a|b)^*$
 $(a+b)^*$
 $(a|b)^*$



$S \rightarrow bS$

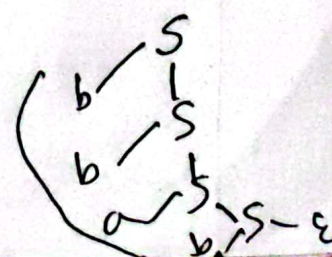
$\rightarrow bbs$

$\rightarrow bbaS$

$\rightarrow bba bS$

$\rightarrow bbab \epsilon$

$\rightarrow bbab$



important note:

(i) a^* ; $s \rightarrow a^s | \epsilon$

(ii) a^+ ; $s \rightarrow a^s | a$

(iii) $(a+tb)^*$; $s \rightarrow a^s | b^s | \epsilon$

Ex

1. Q: Construct CFG for set of all strings which
length at least 2

$$= \{aa, bb, aab, bba, aaab, \dots\}$$

$$\frac{(a+tb)}{A} \frac{(a+tb)}{A} \frac{(a+tb)^*}{B}$$

generate "aaa"

$$S \rightarrow AAB$$

$$A \rightarrow a | b$$

$$B \rightarrow aB | bB | \epsilon$$

$$S \rightarrow AAB$$

$$\rightarrow aAB$$

$$\rightarrow aaB$$

$$\rightarrow aaB$$

$$\rightarrow aaa\epsilon$$

$$\rightarrow aaa$$

2) Construct CFG for set of all strings at least three 0's.

$$= \{ 1010100, 00000, 10001001, \dots \}$$

$A = (0+1)^*$ A^* $\rightarrow$ কোনকথা উল্লি 0.

$S \rightarrow A0A0A0A$

$A \rightarrow 0A \mid 1A \mid \epsilon$

generate: 110010

$S \rightarrow A0A0A0A$

$\rightarrow 1A0A0A0A$

$\rightarrow 11A0A0A0A$

$\rightarrow 11\epsilon A0A0A$

$\rightarrow 1100A0A0A$

$\rightarrow 1100\epsilon A0A0A$

$\rightarrow 1100\epsilon 0A0A$

$\rightarrow 110\epsilon 0A0A$

$\rightarrow 11001A0A$

$\rightarrow 11001\epsilon 0A$

$\rightarrow 110010A \rightarrow 110010\epsilon \rightarrow 110010$

Q=3 Construct CFG for set of all strings which length at most 2. $\Sigma = \{a, b\}$.

$L = \{a, b, ab, ba, \textcircled{aba} \dots\}$.

"अवरोक्त 2 टि" "0-2".

$$\frac{(a+b+\epsilon)}{A} \quad \frac{(A+B+\epsilon)}{A}$$

CFG: $S \rightarrow AA$
 $A \rightarrow a | b | \epsilon$

generate:

$S \rightarrow AA$	$S \rightarrow AA$
$\rightarrow bA$	$\rightarrow aA$
$\rightarrow ba$	$\rightarrow a\epsilon$
	$\rightarrow a.$

Q-4 starts with a and ends with b.

= {ab, aab, abbb, aabb, ...}

CFG:

$S \rightarrow aAb$
$A \rightarrow aA \mid bA \mid \epsilon$

generate: aabb.

$S \rightarrow aAb$

$\rightarrow aaAb$

$\rightarrow aabAb$

$\rightarrow aabb\epsilon b$

$\rightarrow aabb.$

মুঠ ৩৪০ জোড়,
a | | b

ছাফাফে

"a/b/ε" → thodn for
(atb)*.

Q-5: starts and ends with different symbol.

$= \{ab, ba, aaab, baaba, \dots\}$

$a \underbrace{(atb)^*}_A b \mid b \underbrace{(atb)^*}_A a$

CFG:

$S \rightarrow aAb \mid bAb$
 $A \rightarrow aA \mid bA \mid \epsilon$

generate:- aab.

$S \rightarrow aAb$

$\rightarrow aaAb$

$\rightarrow aa\epsilon b$

$\rightarrow aab.$

Q-6: starts and ends with same symbol.

$= \{\epsilon, a, b, aa, bb, aba, bab, \dots\}$

$a \underbrace{(atb)^*}_A a \mid b \underbrace{(atb)^*}_A b \mid a \mid b \mid \epsilon$

CFG:

$S \rightarrow aAa \mid bAb \mid a \mid b \mid \epsilon$
 $A \rightarrow aA \mid bA \mid \epsilon$

generate!

"aaa" $S \rightarrow aAa$

$\rightarrow aaAa$

$\rightarrow aa\epsilon a$

$\rightarrow aaa$

$\rightarrow$

Ambiguity

If a grammar generates the same string in several different ways, we say that the string is derived ambiguously in that grammar. If a grammar generates some strings ambiguously, we say that the grammar is ambiguous.

Ex: CFG: $S \rightarrow stS \mid s*s \mid a$ "ambiguous grammar" pattern: $\rightarrow ata*a$

$S \rightarrow stS$

$\rightarrow at*s$

$\rightarrow ata*s$

$\rightarrow ata*a$

$S \rightarrow s*s$

$\rightarrow st*s$

$\rightarrow at*s$

$\rightarrow ata*s$

$\rightarrow ata*a$

$\downarrow$
"ambiguously generated string"

Remove Null production

$$\textcircled{1} S \rightarrow ABC$$

$$A \rightarrow aA \mid \epsilon$$

$$B \rightarrow bB \mid \epsilon$$

$$C \rightarrow c$$

Removing $A \rightarrow \epsilon$,

$$\downarrow S \rightarrow ABC \mid BC$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid \epsilon$$

$$C \rightarrow c$$

Removing $B \rightarrow \epsilon$,

$$S \rightarrow ABC \mid BC \mid AC \mid C$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid b$$

$$C \rightarrow c$$

* 1

[null production ও ফলাফল " ϵ "।

বাহ্যে দেওয়া হয়।]

[কোনো string ও " ϵ " include থাকলে

তা Remove করতে হয়]

[first step]

[first step থেকে updated

value থেকে শুরু করতে হবে]

$$(2) \quad S \rightarrow ABAC$$

$$A \rightarrow aA \mid \epsilon$$

$$A \rightarrow \epsilon$$

$$B \rightarrow bB \mid \epsilon$$

$$B \rightarrow \epsilon$$

$$C \rightarrow c$$

④ ϵ produce

कपट

Removing $A \rightarrow \epsilon$,

$$S \rightarrow ABAC \mid BAC \mid ABC \mid BC$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid \epsilon$$

$$C \rightarrow c$$

Removing $B \rightarrow \epsilon$

$$S \rightarrow ABAC \mid BAC \mid ABC \mid BC \mid AAC \mid AC \mid C$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid b$$

$$C \rightarrow c$$

special case

Null production का हल "ε" चयन या है,

③ $S \rightarrow aA|A$

$A \rightarrow \epsilon$

$A \rightarrow \epsilon$

$S \rightarrow \epsilon$ [indirectly produce

कहे, cause $S \rightarrow A$ and

$A \rightarrow \epsilon$. So, $S \rightarrow \epsilon$]

Removing $A \rightarrow \epsilon$,

$S \rightarrow aS|\epsilon$

Removing $S \rightarrow \epsilon$,

$S \rightarrow aS|a$

~~R~~

Remove Unit Production from CFG

এখানে একটি variable আরেকটি variable কে produce

করে।

Q:-1

$A \rightarrow AB$

$A \rightarrow a$

$B \rightarrow C \mid b$

$C \rightarrow D$

$D \rightarrow E$

$E \rightarrow a$

$B \rightarrow C \rightarrow D \rightarrow E$

Soln:-

$A \rightarrow AB$

$A \rightarrow a$

$B \rightarrow a \mid b$

$C \rightarrow a$

$D \rightarrow a$

$E \rightarrow a$

note:

[$C \rightarrow D \rightarrow E$ and $E \rightarrow a$
so, $C \rightarrow a$]

[same]

[$D \rightarrow E$ and $E \rightarrow a$ so, $D \rightarrow a$].

Remove unreachable variable:

$A \rightarrow AB$

$A \rightarrow a$

$B \rightarrow a \mid b$

② $S \rightarrow aA \mid B$

$A \rightarrow ba \mid bb$

$B \rightarrow A \mid bba$

$S \rightarrow B$

$B \rightarrow A$

$\therefore S \rightarrow B \rightarrow A$

Soln:- $S \rightarrow aA \mid ba \mid bb \mid bba$

$A \rightarrow ba \mid bb$

$B \rightarrow \text{ba} \mid \text{bb} \mid \text{bba} \} \text{— remove } 20^{\text{th}}$

Remove unreachable variable:

$S \rightarrow aA \mid ba \mid bb \mid bba$

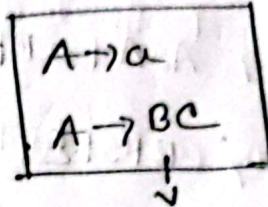
$A \rightarrow ba \mid bb$

(A)

CFG to Chomsky Normal form

CFG $\longleftrightarrow$ (CNF)

① Highest ২টি variable concatenation



আকাঙ্ক্ষা থাকবে,

② একটি terminal থাকবে,

CNF standard form

③ Variable-terminal concatenation আকাঙ্ক্ষা থাকবে না,

Question: 1

$$1. S \rightarrow ASA \mid aB$$

$$A \rightarrow BIS$$

$$B \rightarrow b \mid \epsilon$$

Step: 1 Add $(S_0 \rightarrow S)$.

$$S_0 \rightarrow S$$

$$S \rightarrow ASA \mid aB$$

$$A \rightarrow BIS$$

$$B \rightarrow b \mid \epsilon$$

যদি part গুনো আগে

হতো থাকবে,

Remove

Step-2: 1 Null production:

$$B \rightarrow \epsilon, A \rightarrow \epsilon$$

Remove $B \rightarrow \epsilon$

$S_0 \rightarrow S$

$S \rightarrow ASA|aB|a$

$A \rightarrow B|S|\epsilon$

$B \rightarrow b$

Remove $A \rightarrow \epsilon$

$S_0 \rightarrow S$

$S \rightarrow ASA|aB|a|SA|AS|S$

$A \rightarrow B|S$

$B \rightarrow b$

Step! 3 Remove Unit production:

$S_0 \rightarrow S, S \rightarrow S, A \rightarrow B, A \rightarrow S$

$S_0 \rightarrow ASA|aB|a|SA|AS$

$S \rightarrow ASA|aB|a|SA|AS$

$A \rightarrow b|ASA|aB|a|SA|AS$

$B \rightarrow b$

step-4

Let $X = AS$.

$$S_0 \rightarrow AX | \textcircled{aB} | a | SA | AS.$$

$$S \rightarrow AX | \textcircled{aB} | SA | AS.$$

$$A \rightarrow b | AX | \textcircled{aB} | a | SA | AS.$$

$$B \rightarrow b$$

$$X \rightarrow AS$$

Q

Still rounded figure are not in CNF form. So, we have to convert it into standard form of CNF.

step-5: Let, $Y = a$.

$$S_0 \rightarrow AX | YB | a | SA | AS.$$

$$S \rightarrow AX | YB | a | SA | AS.$$

$$A \rightarrow b | AX | YB | a | SA | AS.$$

$$B \rightarrow b$$

$$X \rightarrow AS.$$

$$\underline{Y} \rightarrow a.$$

Push Down Automata [PDA]

formal Definition :-

A PDA is a 6-tuple $(Q, \Sigma, \Pi, \delta, q_0, F)$.

① Q = finite set of all states.

② Σ = Input Alphabet.

③ Π = Stack Alphabet.

④ $\delta: Q \times \Sigma_{\epsilon} \times \Pi_{\epsilon} \rightarrow P(Q \times \Pi_{\epsilon})$.

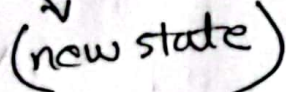
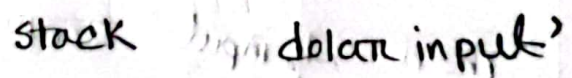
= transition function

⑤ $q_0 \in Q$ = start state.

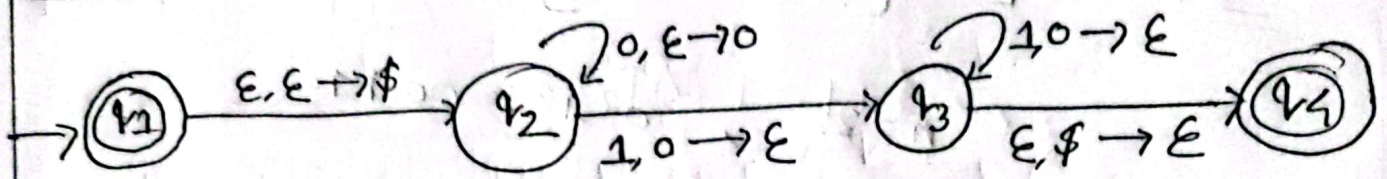
⑥ $F \subseteq Q$ = final state

$L = \{0^n 1^n \mid n \geq 0\}$

(14)



Q:-1 Construct PDA that accepts, $L = \{0^n 1^n \mid n \geq 0\}$.



Explain:-

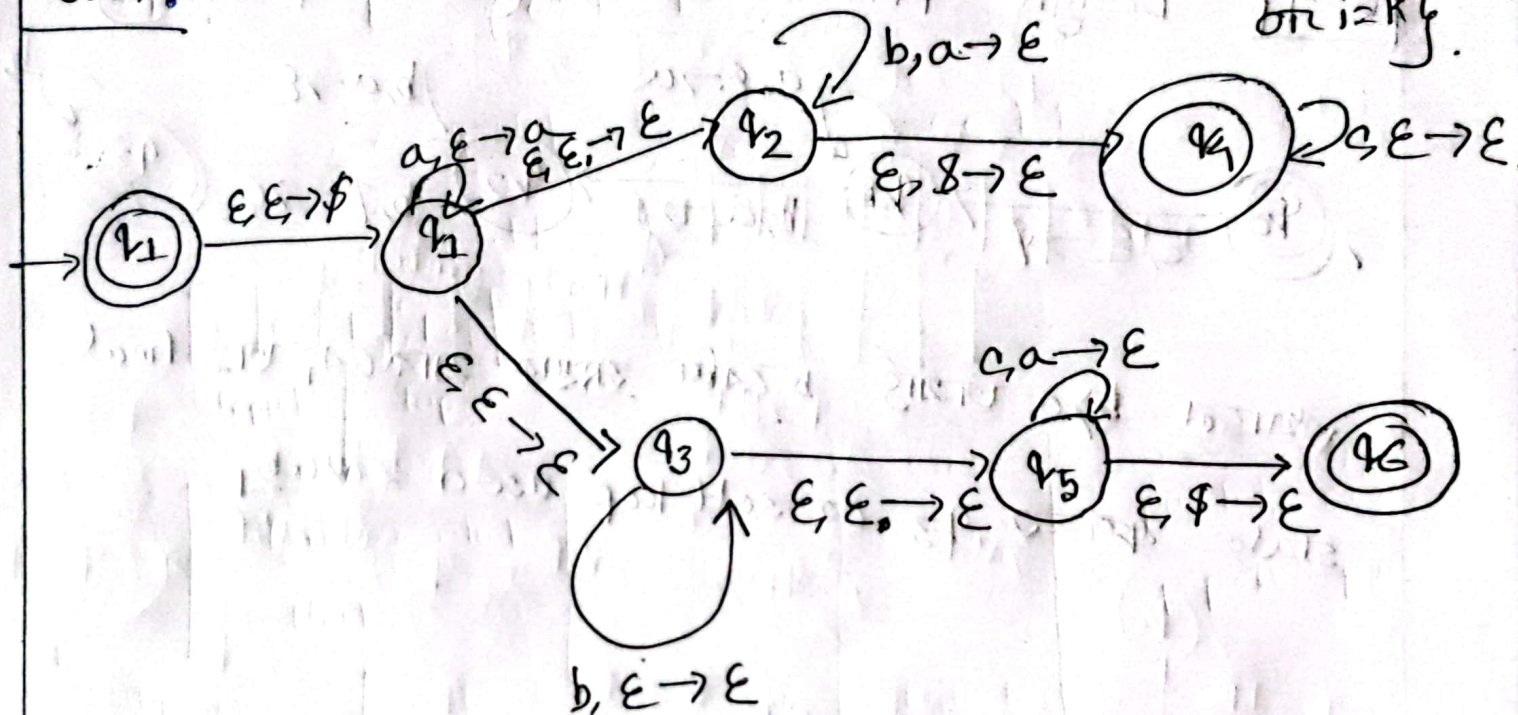
যেহেতু $n \geq 0$. সুতরাং $n=0$ জন্য কোনো কিছু না গেছে
 q_1 final state কিছু input না দিবে অর্থাৎ. no pop (ϵ),
 no push (ϵ) $\rightarrow$ \$ initially pushed হবে, q_2 তে 0 গেলে
 কিছু pop হবে না $\rightarrow$ 0 pushed হবে q_2 state এ,
 Then 1, input হলে, 0 pop হবে $q_2 \rightarrow q_3$ "1" pushed হবে
 q_3 state এ থাকবে, Again input 1 হলে, 0 pop $\rightarrow$ "1" pushed
 হবে, finally noinput 'ε', "\$" popped হবে এবং q_4 হবে
 final state.

state transition:-

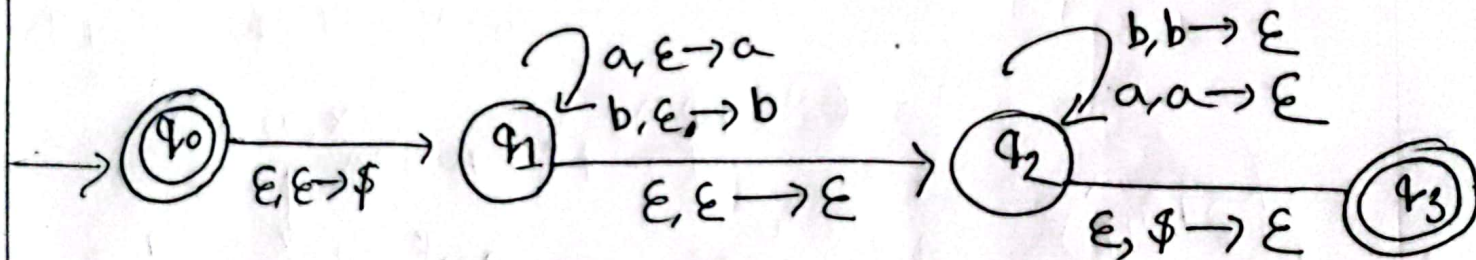
$(q_1, \epsilon, \epsilon) \rightarrow (q_2, \$)$
 $(q_2, 0, \epsilon) \rightarrow (q_2, 0)$
 $(q_2, 1, 0) \rightarrow (q_3, \epsilon)$
 $(q_3, 1, 0) \rightarrow (q_3, 1)$
 $(q_3, \epsilon, \$) \rightarrow (q_4, \epsilon)$



Q-2: Draw PDA for, $L = \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i=j \text{ or } i=k\}$.



Q-3: Draw PDA for, $L = \{ww^R \mid w \in (a,b)^*\}$



ww^R જન્ય (a/b) input for w જરો $[w^R]$ રહના

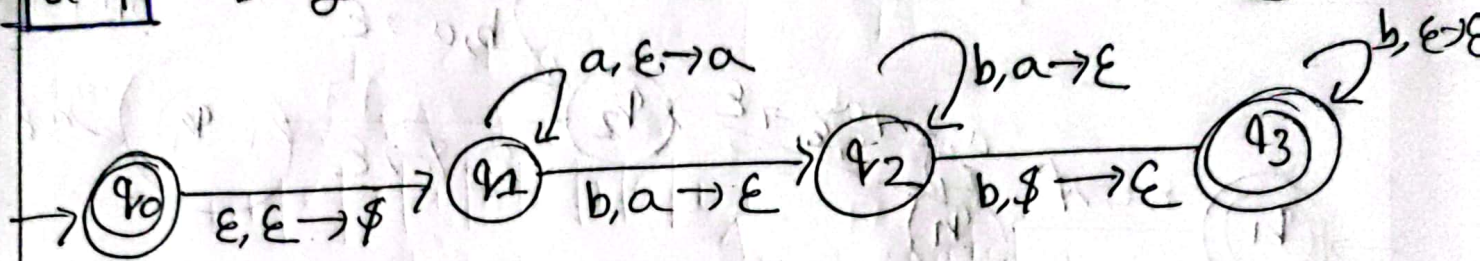
Reverse function જન્ય a input રહના, a pop રહ

→ "ε" pushed, same b જરો જન્ય.

Q-4

Design

PDA for $L = \{ amb^n \mid m \leq n \}$



যখন $b > a$ অর্থাৎ b বেশি অক্ষর থাকবে, তাহলে final state ~~q3~~ $q3$ to self loop used হবে।

Turing Machines

formal Definition:

7-tuple $(Q, \Sigma, \Gamma, \delta, q_0, q_{accept}, q_{reject})$.

1. $Q \rightarrow$ finite set of states.

2. $\Sigma \rightarrow$ Input Alphabet. $\emptyset \neq \Sigma$

3. $\Gamma \rightarrow$ Tape Alphabet. $\Sigma \subseteq \Gamma$

$\emptyset \neq \Gamma$

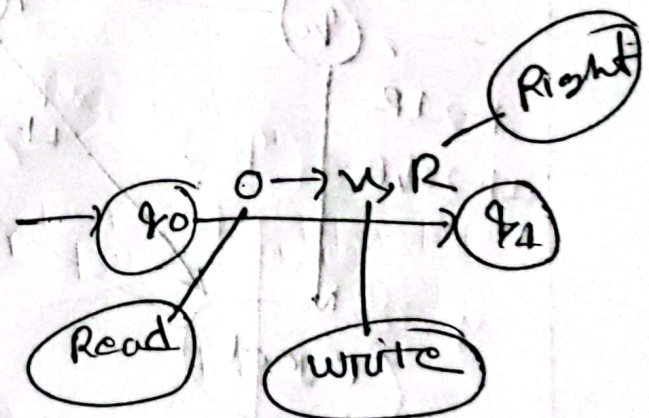
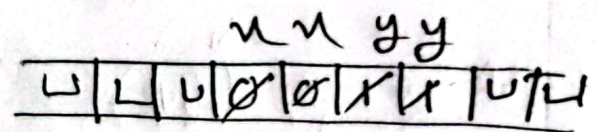
$\forall y \in \Gamma$

4. $\delta: Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$

5. $q_0 \rightarrow$ starting state

6. $q_{accept} \rightarrow$ Accept state

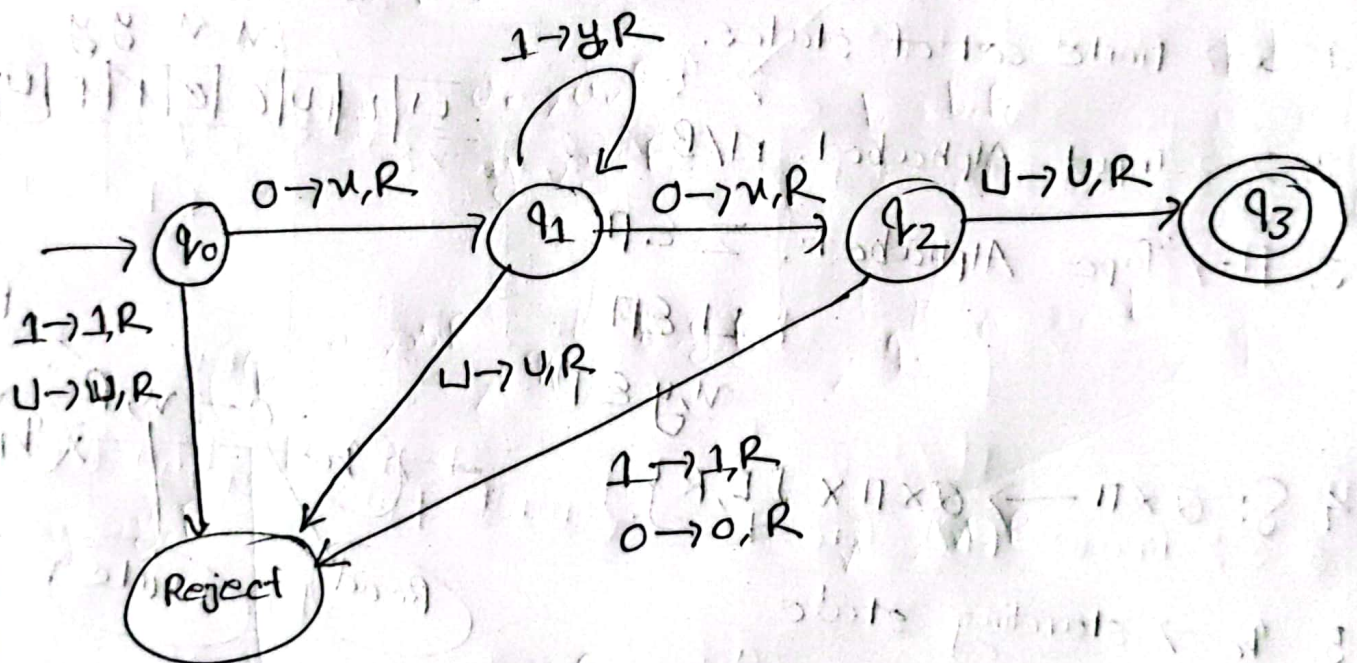
7. $q_{reject} \rightarrow$ Reject state



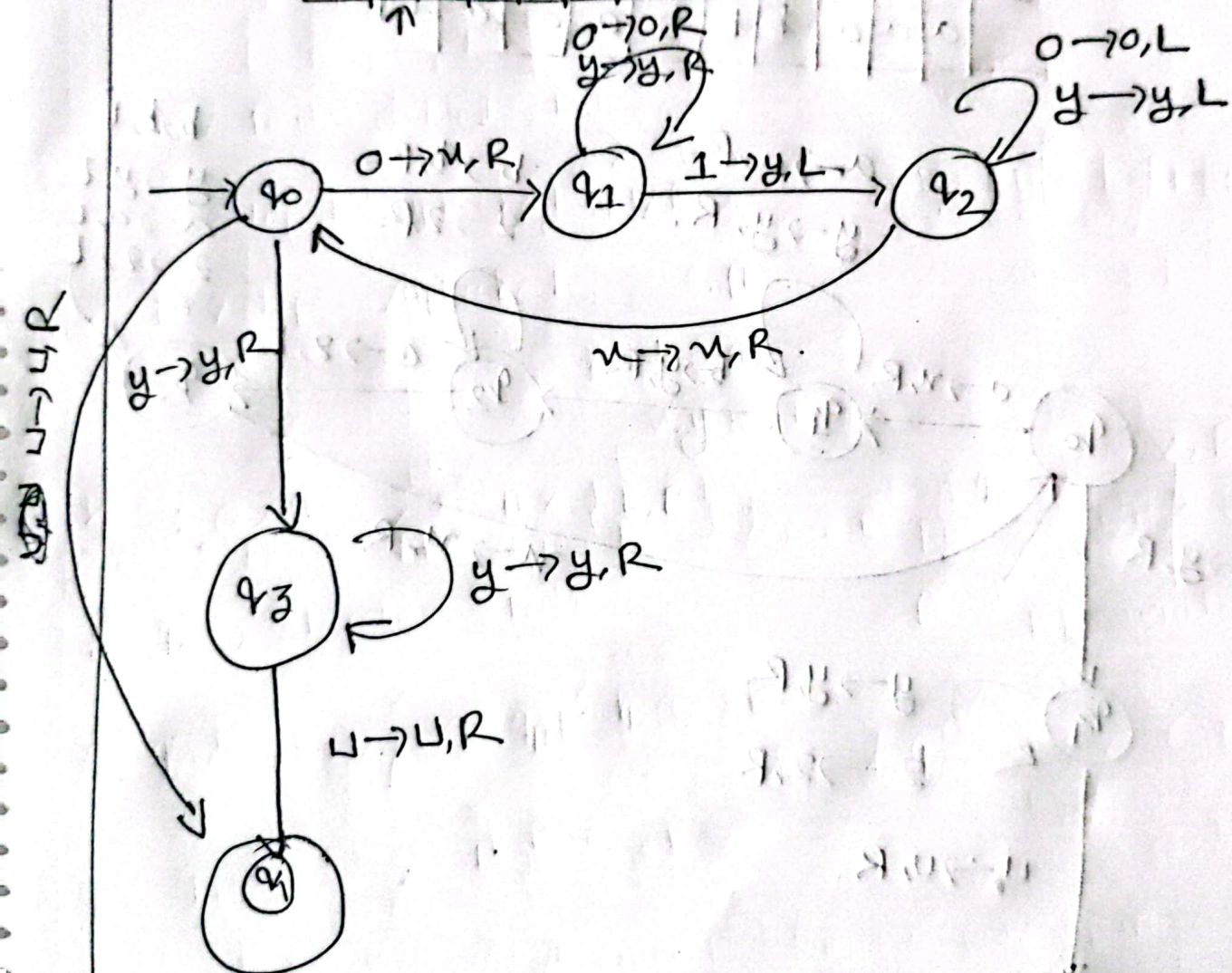
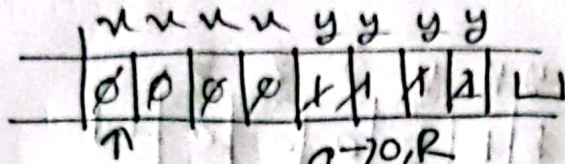
$(q_0, 0) \rightarrow (q_1, u, R)$

$0 \rightarrow u, R \rightarrow$ Right position $\downarrow$ 1 step move
 $\downarrow$
 "0" read
 "0" বদলে u লিখবে

Language. $L = 0.1^* 0.$



Q-2: Design a Turing machine which recognize the language. $L = 0^n 1^n$; $n \geq 0$.



Q-3 Design a machine which recognize the

language. $L = \{anbn^n \mid n \geq 1\}$

